Arista Access Node WHI-ACC-61 Optical Transport Failure SOP (Warning Severity)

Node: WHI-ACC-61 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_2341

Description

This Standard Operating Procedure (SOP) addresses the occurrence of Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) on the Arista Access Node WHI-ACC-61 located at Whitefield East POP. The Arista Access Node WHI-ACC-61 is a critical component of the network infrastructure, responsible for providing a high-speed access interface to the core network. The Optical Transport Failure on this node can be attributed to the degradation of SFP/XFP modules, fiber cuts, or variations in Rx/Tx power levels. This failure domain occurs due to the inherent limitations of optical transmission technology, which is prone to signal degradation and power loss over long distances. The Arista Access Node WHI-ACC-61 uses the Arista EOS (Extensible Operating System) to manage and control the network, which provides a robust and scalable platform for network operation. However, the EOS's reliance on software-based routing and switching protocols can lead to software-related issues, such as misconfigured OSPF routing or faulty BFD (Bidirectional Forwarding Detection) sessions, which can exacerbate the Optical Transport Failure. To mitigate this risk, it is essential to perform regular maintenance and monitoring of the Arista Access Node WHI-ACC-61, including physical checks on the SFP/XFP modules and fiber connections, as well as software-based checks on the EOS configuration and BFD sessions.

Pre-requisite Checks

- Verify that the Whitefield East POP is accessible and that the remote hands team is dispatched to the site.

- Check the redundant link status to ensure that the secondary link is operational and ready to take over in case of a failure.

- Perform a physical inspection of the SFP/XFP modules and fiber connections to identify any signs of damage or degradation.

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP modules	High
Fiber cuts or breaks	Medium
Variations in Rx/Tx power levels	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface e1/1/1 detail, Expected Output: Interface
e1/1/1 is up, line protocol is up, Description: This command is used
```

```
to verify the operational status of the interface, including the SFP/XFP module and fiber connection. The output should indicate that the interface is up and the line protocol is up.

Command: show bfd sessions detail, Expected Output: BFD session is up and running, Description: This command is used to verify the BFD session status, including the session's operational state and any errors that may have occurred. The output should indicate that the BFD session is up and running.

Command: show mpls forwarding-table detail, Expected Output: MPLS forwarding table is populated and up-to-date, Description: This command is used to verify the MPLS forwarding table, including the label switching and forwarding rules. The output should indicate that the MPLS forwarding table is populated and up-to-date.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace module e1/1/1, Description: This step involves physically removing the faulty SFP/XFP module and replacing it with a new one. The Arista Access Node WHI-ACC-61 will automatically detect the new module and bring the interface online.

Step 2: Step: Verify the fiber connection, Command: verify fiber connection e1/1/1, Description: This step involves physically inspecting the fiber connection to ensure that it is secure and not damaged. If the fiber connection is damaged, it will need to be replaced.

Step 3: Step: Reset the BFD session, Command: clear bfd sessions e1/1/1, Description: This step involves resetting the BFD session to clear any errors or issues that may have occurred. The Arista Access Node WHI-ACC-61 will automatically re-establish the BFD session.

Post-Check Verification

```
Command: show interface e1/1/1 detail, Expected Output: Interface e1/1/1 is up, line protocol is up, Description: This command is used to verify that the interface is operational and the SFP/XFP module is functioning correctly.

Command: show bfd sessions detail, Expected Output: BFD session is up and running, Description: This command is used to verify that the BFD session is operational and there are no errors.

Command: show mpls forwarding-table detail, Expected Output: MPLS forwarding table is populated and up-to-date, Description: This command is used to verify that the MPLS forwarding table is populated and up-to-date.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: configure replace original-config, Description: This step involves restoring the original configuration to the Arista Access Node WHI-ACC-61. This will ensure that the node returns to its original operational state.

- Step: Verify the original configuration, Command: show running-config, Description: This step involves verifying that the original configuration was successfully restored. The output should match the original configuration.